MILITARY SPECIFICATION

FUEL CONTROL, MAIN, ELECTROHYDRAULIC

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an electrohydraulic main fuel control used in the gas turbine engine of the M1 tank, referred to herein as "assembly". The assembly is an electrohydraulic device which converts electrical commands to fuel flow and power turbine stator (PTS) inlet guide vane (IGV) actuator flows and pressures. This assembly also contains the PTS and IGV electrical feedback components.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2910

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS
FEDERAL

VV-F-800	- Fuel Oil, Diesel.

MILITARY

MIL-H-6083	- Hydraulic Fluid, Petroleum Base, Preservation and Operations
MIL-C-7024	- Calibrating Fluid, Aircraft Fuel System Components.
MIL-F-13927	- Fungus Resistance Test, Automotive Components.
MIL-P-14232	- Parts, Equipment and Tools for Army Materiel, Packaging of.
MIL-S-22473	- Sealing, Locking and Retaining Compounds; (Single Component).
MIL-C-22520	- Crimping Tools, Terminal, Hand or Power Actuated, Wire Termination and Tool Kits, General Specification for.
MIL-I-23053	- Insulated Sleeving, Electrical, Heat Shrinkable, Flame Retarded, Modified Polyolefin, Flexible, Crosslinked.

STANDARDS
MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-193	- Painting Procedures and Marking for Vehicles, Construction Equipment, and Material Handling Equipment.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic interference Characteristics, Measurement of.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-1460	- Soldering of Electrical Connectors and Printed Wiring Assemblies, Procedure for.
MIL-STD-45662	- Calibration System Requirements-

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.

2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein.

DRAWINGS

12324614	- Fuel Control, Main - Electrohydraulic.
2752412	- Model ED-31A Hydromechanical Controller Test Set.

(Copies of specifications, standards, drawings, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

GENERAL DYNAMICS DEFENSE GROUP

DEOS 2060	- Nuclear Hardening Criteria (U).

(Copies of above document can be obtained from US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS Standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained-

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish assemblies which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the units to be furnished to the Government. All subsequent assemblies delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified herein and in referenced specifications and standards (see 4.8.1).

3.2.1 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.3.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.

3.2.2 Surface treatment. Unless otherwise specified (see 6.2), surfaces other than mounting surfaces shall be treated in accordance with the requirements of MIL-STD-193 (see 4.7.1).

3.3 Design and construction. The assembly shall be designed and constructed as specified herein, and in accordance with Drawing 12324614 and its applicable standards and specifications (see 4.7.2).

3.3.1 Standards of manufacture.

3.3.1.1 Soldering. Unless otherwise specified (see 6.2), soldering of electrical connections shall be in accordance with MIL-STD-454, requirement 5 (see 4.7.1 and 4.7.2).

3.3.1.2 Crimp contacts. Crimp type connector contacts shall be crimped using tools and procedures as specified in MIL-C-22520 (see 4.7.1 and 4.7.2).

3.3.1.3 Heat shrinking. Heat-shrink tubing shall be as specified in MIL-I-23053, and shall be uniformly heat shrunk until the tubing configuration reflects the contour of the surface it is intended to insulate (see 4.7.1 and 4.7.2).

3.3.1.4 Wire stripping. Removal of outer jacket insulation on all wires shall be in accordance with thermal stripping as specified in MIL-STD-1460 (see 4.7.1 and 4.7.2).

3.3.1.5 Thread locking compound. Thread locking compound shall be in accordance with MIL-S-22473 (see 4.7.1 and 4.7~2)-

3.3.1.6 Nuclear hardening. The fuel control assembly shall be designed to withstand the nuclear threat environments as defined in General Dynamics Engineering classified (confidential) document entitled “Nuclear Hardening Criteria (U)”, document control number DEOS 2060 (see 4.7.1 and 4.7.2).

3.3.2 Interchangeability. The assembly shall be interchangeable on existing equipment without modification or adjustments (see 4.7.2).

3.3.3 Weight. The weight of the assembly shall be no greater than 35 pounds (see 4.7.2).

3.4 Performance. Assemblies shall meet the following performance requirements listed below as 3.4.1 through 3.4.11 with their respective tolerances listed in table I. Performance operation shall require diesel fuel conforming to VV-F-800, type DF-2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.

TABLE I. Tolerances.

Function	20°F .856 sp. gr.1/	150°F .800 sp. gr.	85°F .830 sp. gr.
Minimum W_f *2/	43 ± 5 pph	40 ± 5 pph	42 ± 5 pph
0.60 Volts Req *	61 ± 6 pph	58 ± 6 pph	60 ± 5 pph
1.50 Volts Req *	150 ± 12 pph	142 ± 12 pph	147 ± 11 pph
2.00 Volts Req *	199 ± 13 pph	189 ± 13 pph	195 ± 12 pph
4.00 Volts Req *	411 ± 25 pph	390 ± 25 pph	403 ± 17 pph
7.00 Volts Req *	714 ± 28 pph	679 ± 28 pph	700 ± 19 pph
Maximum W_f *	814 ± 30 pph	775 ± 30 pph	799 ± 21 pph
Back-Up W_f	128 ± 11 pph	121 ± 11 pph	125 ± 0 pph
Relief Valve	1125 ± 60 psig	1125 ± 60 psig	1125 ± 45 psig
Cut-Off Valve Leak	7.0 cm^3/m max	7.0 cm^3/m max	5.0 cm^3/m max
PTS F/B @ 5.140"	-6.50 ± .10 volts	-6.50 ± .10 volts	-6.50 ± .05 volts
IGV F/B @ 5.470"	-.67 ± .15 volts	-.67 ± .15 volts	-.67 ± .08 volts
PTS Actuator 0 ma	20 - 160 psid	20 - 160 psid	30 - 150 psid
PTS Actuator 900 ma	20 - 160 psid	20 - 160 psid	30 - 150 psid
IGV Actuator 0 ma	20 - 85 psid	20 - 85 psid	30 - 75 psid
IGV Actuator 900 ma	20 - 85 psid	20 - 85 psid	30 - 75 psid
Press. Test	1100 psig	1100 psig	1100 psig

1/ Sp. gr. = Specific gravity.
2/ W_f = fuel flow.
* Metering Valve Position.
1.0 Volt = 100 pph reference fuel @ 60°F.

NOTES : (1) Reference fuel @ 20°F is 12 centistokes.
(2) Tolerance at 20°F fluid temperature applies at fuel handling unit (fhu) ambient @ -70°F.

3.4.1 Minimum fuel flow. The assembly shall provide a fuel flow of no less than 42 ± 2 pounds per hour (pph) (see 4.7.3.1).

3.4.2 Backup valve operation. The assembly shall Supply a fuel flow of 118 ± 6 pph with loss of electrical power to the assembly (see 4.7.3.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.

3.4.3 Maximum fuel flow. The assembly shall supply a fuel flow of 799 ± 16 pph (see 4.7.3.3).

3.4.4 High pressure relief valve operation. The assembly shall provide a pressure relief valve to limit the Pressure at the control discharge port to 1150 ± 25 pounds per square inch gage (psig) (see 4.7.3.4).

3.4.5 Fuel flow schedule. The assembly shall schedule fuel flow in accordance with figure 1, with null current set to 275 ± 5 milliamperes (ma) at 0.144 volts root mean square (vrms) (see 4.7.3.5).

3.4.6 Cutoff valve and check valve operation. The assembly shall provide a combination cutoff and fuel flow check valve. The cutoff valve shall be normally open and energized closed with leakage not to exceed 5.0 cubic centimeter per minute (cm /m) at 920 ± 10 psig. The check valve shall open at 54 ± 5 psig with leakage not to exceed 0.25 cm^3/m at 40 to 42 psig (see 4.7.3.6).

3.4.7 Power turbine stator (PTS) feedback voltage output l The assembly shall provide an electrical feedback signal proportional to the positioning of the PTS feedback lever. The position shall be 5.135 - 5.145 inches in accordance with Drawing 12286971. The relationship of the PTS secondary voltage versus this dimension, for an excitation voltage of 5.773 vrms on the primary, shall be as follows:

Specified dimension	PTS secondary voltage
5.140 ± .001 inch	1.2499 ± .0192 vrms
6.122 ± .001 inch	.1192 ± .0577 vrms

For specified pin definition, see figure 2 (see 4.7.3.7).

3.4.8 Inlet guide vane (IGV) feedback voltage output. The assembly shall provide an electrical feedback signal proportional to the positioning of the IGV feedback lever. The position shall be 5.465 - 5.475 inches in accordance with Drawing 12286971. The relationship of the IGV secondary voltage versus this dimension for an excitation voltage of 5.773 vrms on the primary shall be as follows:

Specified dimension	IGV secondary voltage
5.470 ± .001 inch	.0769 ± .0115 vrms
7.040 ± .001 inch	1.2302 ± .0346 vrms

For specified pin definition, see figure 2 (see 4.7.3.8).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.

3.4.9 PTS actuator flow. The assembly shall provide a relative flow between ports 7 and 8, in accordance with Drawing 12286971, as a function of electrical current to the PTS solenoid. The relationship of electrical current to the flow shall be as follows (see 4.7.3.9):

PTS solenoid current	Pressure/flow requirements (350 pound per square inch differential (psid) P_1 - P_0 across the fuel handling unit (fhu).
(Pins g and h)	
0 (+10, -0) ma	When port 7 is 24-150 psid greater than port 8, actuator flow shall be 220-570 pounds per hour (pph).
350 ma ± 50 ma	When port 7 is equal to port 8, actuator flow shall be 0 pph and pressure shall be equal.
900 ma ± 5 ma	When port 8 is 24-150 psid greater than port 7, actuator flow shall be 220-570 pph.

3.4.10 IGV actuator flow. The assembly shall provide a relative flow between ports 5 and 6, in accordance with Drawing 12286971, as a function of electrical current to the IGV solenoid. The relationship of electrical current to the flow shall be as follows (see 4.7.3.10):

IGV solenoid current	Pressure/flow requirements
	350 psid, P_1 - PO across the fhu.
(Pins a and b)	
o (+ 10, -0) ma	When port 6 is 15-75 psid greater than port 5, actuator flow shall be 90-210 psid.
350 ma ± 50 ma	When port 6 is equal to port 5, actuator flow shall be 0 pph and pressure shall be equal.
900 ma ± 5 ma	When port 5 is 15-75 psid greater than port 6, actuator flow shall be 90-210 pph.

3.4.11 Pressure resistance. The assembly shall withstand a pressure test of 1100 psig with no leakage (see 4.7.3.11 and 6.3.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.

3.5 Environmental. The assembly shall meet the operational performance requirements specified in 3.4 during and after, as applicable, the following environmental conditions of 3.5.1 through 3.5.9.

3.5.1 High temperature. The assembly shall operate as specified in 3.4 after exposure to an ambient temperature of 250 + 10°F, with an inlet fuel temperature of 150 + 10°F (see 4.7.4.1).

3.5.2 Low temperature. The assembly shall operate as specified in 3.4 and achieve 10 simulated starts during and after exposure to an ambient temperature of -60 ± 10°F with fuel viscosity ranging from 4.1 to 5.3 centistokes (see 4.7.4.2).

3.5.3 Gun firing shock resistance. The assembly shall operate as specified in 3.4 during and after exposure to shocks of 55 + 5.5g for 2.5 + 0.2 milliseconds (ins) duration half sine wave in each direction of three mutually perpendicular axes (see 4.7.4.3).

3.5.4 Ballistic shock resistance. The assembly shall operate as specified in 3.4 after subjection to shocks of 200 ± 20 gravity units (g) for 0.5 + 0.1 ms duration half sine wave in each direction of three mutually perpendicular axes (see 4.7.4.4).

3.5.5 Basic shock resistance The assembly shall operate as specified in 3.4 during and after exposure to shocks of 20 ± 2 g for 18 ms duration half sine wave in each direction of three mutually perpendicular axes (see 4.7.4.5).

3.5.6 Vibration resistance. The assembly shall operate as specified in 3.4 during and after subjection to the peak vibration envelope of figure 3 in each of three mutually perpendicular axes. The vibration shall be imposed at a logarithmic sweep rate of 20 minutes per sweep cycle from 5 to 500 to 5 Hertz (Hz) (see 4.7.4.6).

3.5.7 Submergence. The assembly shall operate as specified in 3.4.9 and 3.4.10 to regulate the required fuel flow rate and the flow and pressure to the PTS (3.4.9) and IGV (3.4.10) servo valves when subjected to a water submergence of 6 feet (see 4.7.4.7).

3.5.8 Sand and dust. The assembly shall operate as specified in 3.4 during and after exposure to a sand and dust environment (see 4.7.4.8).

3.5.9 Fungus. The assembly shall operate as specified in 3.4 after exposure to conditions favorable to fungus growth. Unless otherwise specified (see 6.2), these conditions shall be as defined in MIL-F-13927 (see 4.7.4.9).

3.5.10 Electromagnetic interference and compatibility. The assembly shall be capable of meeting the emission and susceptibility requirements of MIL-STD-461, notice 4, tests CE01, CE03, CE04, CE05, CS01, CS02, CS06, RE02 and RS03/RS03.1 (see 4.7.4.10).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.

3.6 Endurance and reliability.

3.6.1 Endurance. The endurance of the assembly shall be based on satisfying the performance requirements of 3.4 after 350 hours of operation (see 4.7.4.11 and 6.2) in accordance with the fuel flow schedule of figure 1 and the duty cycle of figure 4.

3.6.2 Reliability. The reliability of the assembly shall be demonstrated based on achieving the mean time between failures in hours (MHBF) listed below upon completion of endurance operations (see 4.7.4.11.2 and 6.2). The MHBFs shall be calculated by dividing the total endurance operating hours by the number of reliability failures defined below:

a. 10,000 MHBF for failures causing loss of total fuel schedule.

b. 4,800 MHBF for failures causing fuel restriction resulting in a major fuel schedule change, defined as a 20% loss of fuel flow [564-600 pounds per hour (pph)] at rated power.

c. 1,200 MHBF for failures causing unscheduled maintenance action.

3.7 Identification marking. Identification marking shall be in accordance with MIL-STD-130 and as noted on Drawing 12286971 (see 4.7.2).

3.8 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9 (see 4.7.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor’s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
   1. Preproduction inspection (see 4.4.1).
   2. Initial production inspection (see 4.4.2).

b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Tests (see 4.5.3).

c. Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the environmental test conditions as follows:

Fuel temperature	85 ± 5°F
Air temperature	73 ± 18°F
Barometric pressure	25.5 to 30.5 inches of mercury (Hg)
Relative humidity	50 ± 30 percent

4.3.1 Test equipment. The following test equipment shall be used in the performance of the examinations and tests specified herein:

a. Test set, Bendix model EK-31A or equivalent.
b. Fixture assembly, Bendix part number 2524777-G1 or equivalent.

4.4 First article inspection. First article inspection shall be performed on preproduction or initial production samples as specified herein. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply assemblies that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of two assemblies. Preproduction inspection shall consist of the inspections in the order listed and as specified in table II, except testing for endurance and reliability. When specified (see 6.2), endurance and reliability test will be conducted on preproduction samples, as specified in 6.2. Samples undergoing endurance and reliability testing shall not be subjected to environmental tests of 4.7.4.1 through 4.7.4.9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.

4.4.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select two assemblies of the first ten assemblies produced under the production contract for initial production inspection. Initial production units shall be inspected in the order listed and as specified in table II.

4.4.3 First article inspection failure. Deficiencies found during, or as a result of, first article inspection shall be cause for rejection of the first article sample until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, first article inspection shall be evidence that all items already produced parior to completion of first article inspection are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government will not accept products until first article inspection is completed to the satisfaction of the Government.

TABLE II. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance		Control
				Examination	Tests	
Materials and construction	3.2, 3.2.2, 3.3.1.1 thru 3.3.1.6	4.7.1	X	X		
Defects (see 4.5.2 and table III)	3.3. thru 3.3.3, 3.7 and 3.8	4.7.2	X	X		
Minimum fuel flow	3.4.1	4.7.3.1	X		X	
Backup valve operation	3.4.2	4.7.3.2	X		X	
Maximum fuel flow	3.4.3	4.7.3.3	X		X	
High pressure relief valve operation	3.4.4	4.7.3.4	X		X	
Fuel flow schedule	3.4.5	4.7.3.5	X		X	
Cutoff valve and check valve operation	3.4.6	4.7.3.6	X		X	
PTS feedback voltage output	3.4.7	4.7.3.7	X		X	
IGV feedback voltage output	3.4.8	4.7.3.8	X		X	
PTS actuator flow	3.4.9	4.7.3.9	X		X	
IGV actuator flow	3.4.10	4.7.3.10	X		X	
Pressure resistance	3.4.11	4.7.3.11	X		X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.

TABLE II. Classification of inspections - Continued.

Title	Requirement	Inspection	First article	Quality conformance		
				Examination	Tests	Control
High temperature	3.5.1	4.7.4.1	X			X
Low temperature	3.5.2	4.7.4.2	X			X
Gun firing shock resistance	3.5.3	4.7.4.3.1	X			
		4.7.4.3.2				X
Ballistic shock resistance	3.5.4	4.7.4.4	X 1/			
Basic shock resistance	3.5.5	4.7.4.5.1	X			
		4.7.4.5.2				X
Vibration resistance	3.5.6	4.7.4.6.1	X			
		4.7.4.6.2				X
Submergence	3.5.7	4.7.4.7	X			X
Sand and dust	3.5.8	4.7.4.8	X			X
Fungus	3.5.9	4.7.4.9	X			X
Electromagnetic interference and compatibility	3.5.10	4.7.4.10	X 1/			
Endurance	3.6.1	4.7.4.11.1	X 2/			
Reliability	3.6.2	4.7.4.11.2	X 2/			

1/ Preproduction only.
2/ Samples shall not be subjected to tests 4.7.4.1 through 4.7.4.9.

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the assemblies of one type and part numbers, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105. Before sampling may be initiated, the contractor shall establish by examination of at least 20 consecutively produced assemblies that the process average percent defective, as defined in MIL-STD-105, is not greater than the specified AQLs. If the computed process average for the requirements exceed the specified AQL, 100% inspection shall be continued until the process average for 20 consecutive assemblies is less than the specified AQL.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.

4.5.2 Examination.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4,5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL).

Classification	AQL
Major	1.0
Minor	4.0

4.5.2.3 Classification of defects. For examination purposes, defects shall be classified as listed in table III.

TABLE III. Classification of defects.

Category	Defect	Method of inspection
Critical	None	
Major	AQL 1.0% Defective	
101	Dimensions affecting interchangeability not within tolerance (see 3.3 and 3.3.2).	SIE 1/
102	Improper materials (see 3.3.1.3 and 3.3.I.5).	Visual
103	Improper weight (see 3.3.3).	Scale
104	Faulty workmanship affecting performance (see 3.3.1.1 and 3.3.1.6).	Visual
Minor	AQL 4.0% Defective	
201	Dimensions not affecting interchangeability not within tolerance (see 3.3).	SIE
202	Noncoformance in design and construction (see 3.3.1.2, 3.3.1.3 and 3.3.1.4).	Visual
203	Improper marking (see 3.7).	Visual
204	Faulty workmanship affecting appearance (see 3.8).	Visual

1/ SIE = Standard Inspection Equipment

4.5.3 Test. Each assembly shall be subjected to the quality conformance tests specified in table II.

4.5.3.1 Failure. Assemblies rejected as a result of testing specified in 4.5.3 may be repaired or reworked and resubmitted for examination and testing.

4.6 Control tests. Unless otherwise specified (see 6.2), control tests shall be conducted by the supplier for maintaining control of the requirements whose length, expense, or destructiveness prohibit their inclusion under lot-by-lot inspection. Control test assemblies shall be inspected for the characteristics specified in Table III. These tests need not be performed on the first control inspection quantity when initial production inspection has been satisfactorily accomplished.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.

4.6.1 Control tests at one in two-hundred (1/200). Unless otherwise specified (see 6.2), control tests shall be performed on one assembly from each lot of 200 assemblies consecutively produced except that not more than one test shall be performed in a 3-month period nor less than one test in a 6-month period.

4.6.2 Control test frequencies other than one in two hundred. If specified (see 6.2), control inspection shall be performed on 1 assembly from 10. However, not more than 1 test shall be performed in a 6-month period nor less than 1 test shall be performed in a 2-month period period. The assembly may be selected from either the first 10 assemblies produced for the interval defined, or the last 10 assemblies produced from the preceding interval.

4.6.3 Failure.

4.6.3.1 Control test failure of one in two hundred. Failure of the assembly to meet the specified control tests may be considered cause for rejection of the production quantity represented. When cause for failure is determined, necessary corrections shall be made and proven by testing three randomly selected assemblies from that production quantity. One of three assemblies shall be subjected to all specified tests. Control test verification for the two remaining assemblies shall be limited to the parameters directly related to the failure cause and the parameters affected by the corrective actions taken.

4.6.3.2 Control test failure of other than one in two hundred. Failure of a control assembly to meet the specified tests may be considered cause to reject subsequent assemblies until deficiencies that caused the failure are corrected and proven by testing three consecutively produced assemblies. Control inspection verification for the assemblies shall be limited to the parameters directly related to the failure cause and the parameters affected by the corrective actions taken.

4.7 Methods of inspection.

4.7.1 Materials and construction. Conformance to 3.2, 3.2.2, 3.3.1.1 through 3.3.1.6 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.7.2 Defects. Conformance to 3.3 through 3.3.3, 3.7 and 3.8 shall be determined by examination for the defects listed in table II. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.

4.7.3 Performance tests.

4.7.3.1 Minimum fuel flow. To determine conformance to 3.4.1, the fuel control assembly shall be tested using test equipment specified in 4.3.1 and type II calibrating fluid as specified in MIL-C-7024.

4.7.3.2 Backup valve operation. To determine conformance to 3.4.2, electrical power to the assembly shall be off and the backup fuel flow shall be measured using the test equipment specified in 4.3.1 and type II calibrating fluid as specified in MIL-C-7024.

4.7.3.3 Maximum fuel flow. To determine conformance to 3.4.3, the fuel flow shall be meaured using test equipment specified in 4.3.1 and type II calibrating fluid as specified in MIL-C-7024.

4.7.3.4 High pressure relief valve. To determine conformance to 3.4.4, the assembly shall be visually examined to verify the provision of a relief valve. The assembly shall be tested using equipment specified in 4.3.1 and type II calibrating fluid as specified in MIL-C-7024 to measure pressure at the control discharge port.

4.7.3.5 Fuel flow schedule. To determine conformance to 3.4.5, the assembly scheduled fuel flow shall be measured using test equipment specified in 4.3.1 and type II calibrating fluid as specified in MIL-C-7024.

4.7.3.6 Cutoff valve and check valve operation. To determine conformance to 3.4.6, the assembly shall be examined to verify the provision of a combination cutoff a fuel flow check valve. The assembly shall be tested for opening, closing and leakage with equipment specified in 4.3.1 and type 11 calibrating fluid as specified in MIL-C-7024.

4.7.3.7 PTS feedback voltage output. To determine conformance to 3.4.7, the position, specified dimension and voltage shall be measured using test equipment specified in 4.3.1 and type 11 calibrating fluid as specified in MIL-C-7024.

4.7.3.8 IGV feedback voltage output. To determine conformance to 3.4.8, the position, specified dimension and voltage shall be measured using test equipment specified in 4.3.1 and type 11 calibrating fluid as specified in MIL-C-7024.

4.7.3.9 PTS actuator flow. To determine conformance to 3.4.9, the relationship of the voltage to the flow shall be measured using test equipment specified in 4.3.1 and type II calibrating fluid conforming to MIL-C-7024.

4.7.3.10 IGV actuator flow. To determine conformance to 3.4.10, the relationship of the voltage to the flow shall be measured using test equipment specified in 4.3.1 and type II calibrating fluid conforming to MIL-C-7024.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.

4.7.3.11 Pressure resistance. To determine conformance to 3.4.11, the assembly shall be tested to withstand 1100 psig with no leakage (see 6.3.3), using test equipment specified in 4.3.1 and type II calibrating fluid conforming to MIL-C-7024.

4.7.4 Environmental. The assembly shall meet the applicable a or b environmental conditions; all assembles shall conform to c environmental condition:

a. Assemblies undergoing first article Inspections shall meet the operational performance requirements of 3.4 during and after, as applicable, the environment testing specified in 4.7.4.1 through 4.7.4.9.

b. Assemblies undergoing control testing shall meet the operational performance requirements only once after all environmental tests, as specified, have been performed.

c. Recalibration shall not be performed during or after subjection to environmental testing.

4.7.4.1 High temperature. To determine conformance to 3.5.1, the assembly shall be mounted in a temperature chamber in operational condition on a base whose temperature shall be maintained at 300 ± 20°F. Ambient room temperature shall be maintained at 250 ± 10°F and the DF02 (VV-F-800) fuel shall be maintained at an inlet temperature of 150 ± 10°F. The fuel Shall be contaminated with water to a level of 0.01% by volume.

a. The assembly shall be subjected to 8 hours of duty cycle, as defined in figure 4, while maintained at the above specified conditions. The assembly shall meet the performance requirements of 3.4 while operating under these conditions.

b. The assembly shall be returned to standard ambient conditions specified In 4.3, nonoperating, and stabilized.

c. The assembly shall then be operated under standard ambient conditions and meet performance requirements specified in 3.4.

4.7.4.2 Low temperature. To determine conformance to 3.5.2, the assembly shall be mounted in a temperature chamber in operational condition. The ambient air temperature shall be -60 ± 10°F with fuel viscosity ranging from 4.1 to 5.3 centistokes.

a. The assembly shall temperature soak for 8 hours at the conditions specified above. The assembly shall achieve 10 simulated starts under these conditions.

b. The assembly shall be returned to standard ambient conditions specified in 4.3 and stabilized. The assembly shall then be operated under these conditions and shall again achieve 10 simulated starts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.

4.7.4.3 Gun firing shock resistance.

4.7.4.3.1 Gun firing shock resistance for first article testing. To determine conformance to 3.5.3, the assembly shall be mounted, using the installed interface configuration, and in operational condition. A total of three half sine wave shock pulses shall be applied in both directions along each of three mutually perpendicular axes (total of 18 shocks). Peak amplitude shall be 55 ± 5.5g for duration of 2.5 ± 0.2 ms. During shock exposure the assembly shall be operating and shall be monitored to verify conformance to the operating performance requirements of 3.4. Upon completion of the shock test, the assembly shall again be inspected to meet the operational performance of 3.4.

4.7.4.3.2 Gun firing shock resistance for control testing. to determine conformance to 3.5.3, the assembly shall be subjected to the test of 4.7.4.3.1, except the shock pulses shall be 40 ± 4g for 2.5 ± 0.2 ma.

4.7.4.4 Ballistic shock resistance. To determine conformance to 3.5.4, the assembly shall be mounted using the installed interface configuration in a nonoperational and pressurized mode. The assembly shall be subjected to three half sine wave shocks of 200 ± 20g for a duration of 0.5 ± 0.1 ma in each direction along each of three mutually perpendicular axes (total of 18 shocks). Upon completion of the shock test, the assembly shall be inspected to meet the operational performance requirements 3.4.

4.7.4.5 Basic shock resistance.

4.7.4.5.1 Basic shock resistance for first article testing. To determine conformance to 3.5.5, the assembly shall be mounted in operational condition using the installed interface configuration. A total of three half sine wave shock pulses of 20 ± 2g shall be applied in both directions along each of three mutually perpendicular axes for a duration of 18 ms. During exposure, the assembly shall be operating and shall be monitored to verify conformance to the operating performance requirements of 3.4. Upon completion of the shock test, the assembly shall again be inspected to meet the operational performance of 3.4.

4.7.4.5.2 Basic shock resistance for control testing. To determine conformance to 3.5.5, the assembly shall be subjected to the teat of 4.7.4.5.1, except the shock pulse shall be 15 ± 1.5g for a duration of 18 ma.

4.7.4.6 ~~Vibration resistance.~~

4.7.4.6.1 Vibration resistance for first article testing. To determine conformance to 3.5.6, the assembly shall be mounted in operational condition using the installed interface configuration. Sinusoidal vibration tests shall be conducted over a period of 6 hours. During this test the assembly shall be operating and the following test conditions shall be applied:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.

a. The vibration curve shall be in accordance with figure 3.

b. The ambient room temperature shall be 140°F.

c. The frequency of applied vibration shall be 5 to 500 to 5 Hz within a logarithmic resonant sweep time of 20 minutes. The sweep shall be conducted in each of the three mutually perpendicular axes, to identify up to four of the most severe resonant frequencies. A resonant dwell shall then be conducted for 20 minutes at each of the four resonant frequencies, for a total of 120 minutes in each axis. If less than four resonant frequencies are found, the balance of the 120 minute vibration time in each axis shall be spent cycling the assembly over the specified frequency range.

d. During exposure, the assembly shall be operating and shall be monitored to verify conformance to the operating performance requirements of 3.4. Upon completion of the vibration test, the assembly shall again be inspected to meet the operational performance requirements of 3.4.

4.7.4.6.2 Vibration resistance for control test. To determine conformance to 3.5.6, the assembly shall be subjected to the test of 4.7.4.6.1, except the the total test time in each axis shall be 75 minutes.

4.7.4.7 Submergence. To determine conformance to 3.5.7, all fluid and electrical fittings shall be capped and the assembly shall be subjected to a one-hour cycle test consisting of immersion in 70°f water from a depth of 6 feet to 8 feet at a rate of 4 cycles per minute. Within two hours after completion, the calibration shall be checked to verify conformance of fuel flow and the flow of pressure.

4.7.4.8 Sand anti dust. To determine conformance to 3.5.8, the fuel control assembly (with the PTS and IGV feedback assemblies installed and rigged for cycling stop to stop) shall be subjected to a sand and duet environment for a period of 10 hours. All other body portions and openings shall be capped. Prior to test, 5 ounces of MIL-H-6083 hydraulic fluid shall be poured over each feedback mechanism. Both lever assemblies shall be rigged to cycle from full open to full closed very 30 seconds, with a 2 degree dither cycle at a rate of 2 cycles per second superimposed for a 10-second interval. A stream of dust-laden air 0.1-0.5 gram per cubic foot density shall be directed at the feedback assemblies at a velocity of 600-1200 feet per minute. Test chamber temperature and relative humidity of 80 $\pm$ 10°F and 10 percent maximum, respectively, shall be maintained during the test of 10 hours duration. Lever running torques and static feedback voltage readings of several positions shall be recorded before, during,and after tests.

Source: https://assist.dla.mil -- Downloaded: 202610-23T01:41Z
Check the source to verify that this is the current version before use.

4.7.4.9 Fungus. To determine conformance to 3.5.9, the fuel control assembly shall be tested in accordance with method 508.3 of MIL-STD-810. In lieu of fungus testing, certification (4.8.1) with supporting data, may be provided attesting that the assembly is constructed of materials that would not support fungus growth.

4.7.4.10 Electromagnetic interference and compatibility. To determine conformance to 3.5.10, the fuel control assembly shall be tested in accordance with MIL-STD-462, notice 3.

4.7.4.11 Endurance and reliability. To demonstrate conformance to 3.6.1 and 3.6.2, the assembly shall be operated using the test equipment specified in 4.3.1 or equivalent, and in accordance with the fuel flow schedule and duty cycle specified in 3.6.1.

4.7.4.11.1 Endurance test. To demonstrate conformance to 3.6.1, the assembly shall be operated for a total of not less than 350 hours. During testing-, reliability failures (see 3.6.2) and other deficiencies shall be recorded. Repair/replacement of failed items shall be made in order to complete 361.4 hours of operation. Upon completion of the endurance operating hours, the performance tests of 3.4 shall be conducted.

4.7.4.11.2 Reliability verification. To determine conformance to 3.6.2, the reliability of the assembly, in MHBF, shall be calculated by multiplying the endurance operating hours by the number of samples and dividing the result by the number of reliability failures recorded under 4.7.4.11.1

5. PACKAGING

5.1 Preservation, packaging, packing, and marking Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The fuel control assembly is to be used in the fuel management system on the AGT 1500 turbine engine.

6.2 Ordering data. Acquisition documents should specify the following:

a. ~~Title, number, and date of this specification.~~
b. If first article inspection is not required (ace 3.1).
c. If surface treatment shall be other than as specified (see 3.2.2).
d. If endurance and reliability testing is required (see 3.6.1 and 4.4.1).
e. Sample size for endurance and reliability testing (see 3.6.1 and 4.4.1).
f. If soldering of electrical connections shall be other than as specified (see 3.3.1.1).

Source: https://assist.dla.mil -- Downloaded: 20261-23T01:41Z
Check the source to verify that this is the current version before use.

g. If conditions for fungus growth shall be other than as specified (see 3.5.9).
h. If responsibility for inspection shall be other than specified (see 4.1).
i. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
j. If inspection conditions shall be other than as specified (see 4.3).
k. If preproduction samples are required (see 4.4.1).
l. If preproduction samples shall be other than as specified (see 4.4.2).
m. If control tests are not required (see 4.6).
n. If control test frequency shall be other than one assembly from each lot of 200 (see 4.6.1 and 4.6.2).
o. Selection of applicable level and packaging requirements (see 5.1).

6.3 Definitions.

6.3.1 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.3.2).

6.3.2 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.3.3 Leak. Loss of fluid which forms more than one drop per hour.

6.4 Subject term (key word) listing.

Electrohydraulic main fuel control.
Fuel, electrohydraulic, main control.

6.5 Supersession data. This specification supersedes AVCO Lycoming Product Fabrication Specification, E2159B, dated 26 April 1983.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.

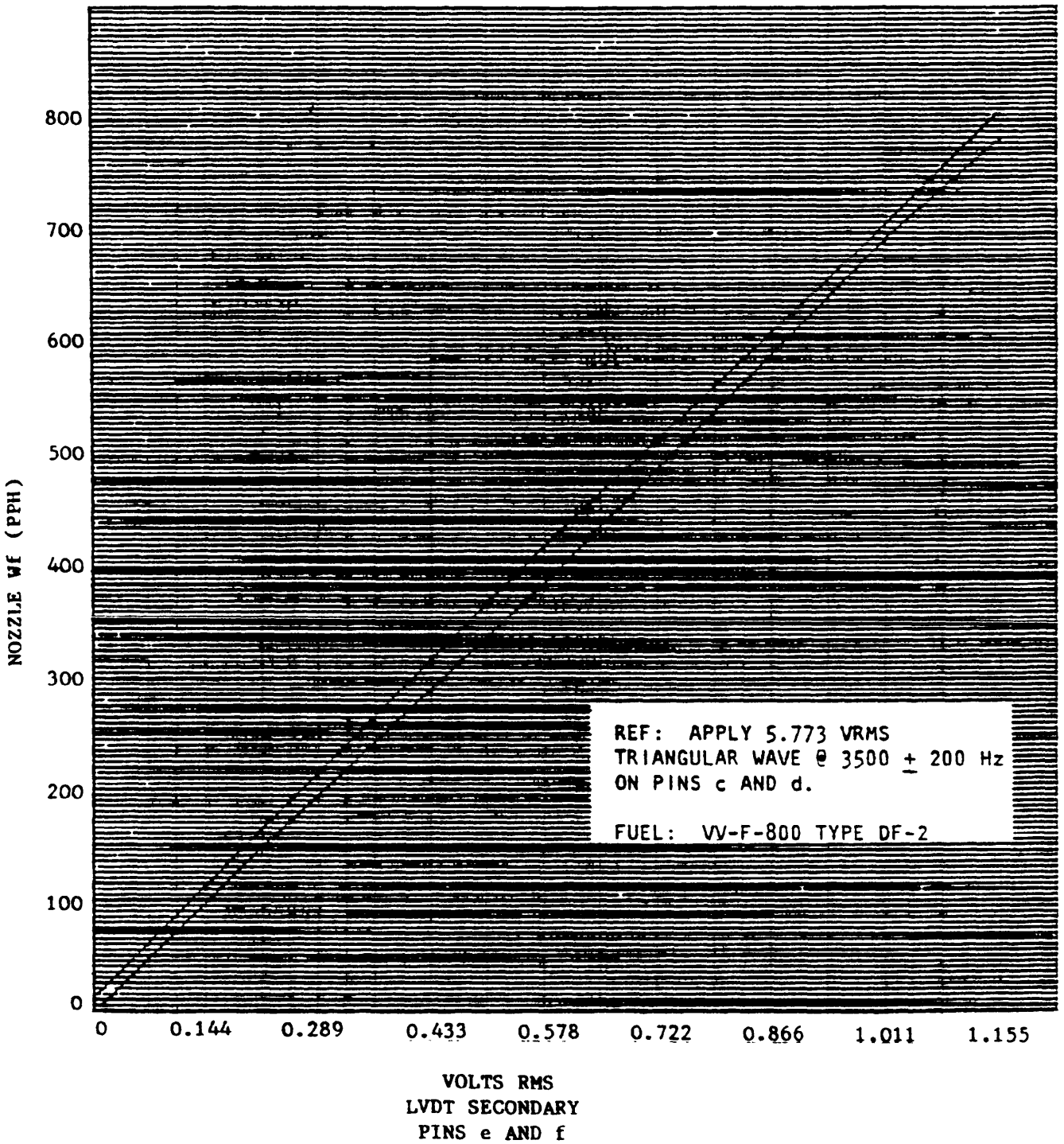


FIGURE 1. Fuel flow schedule.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.

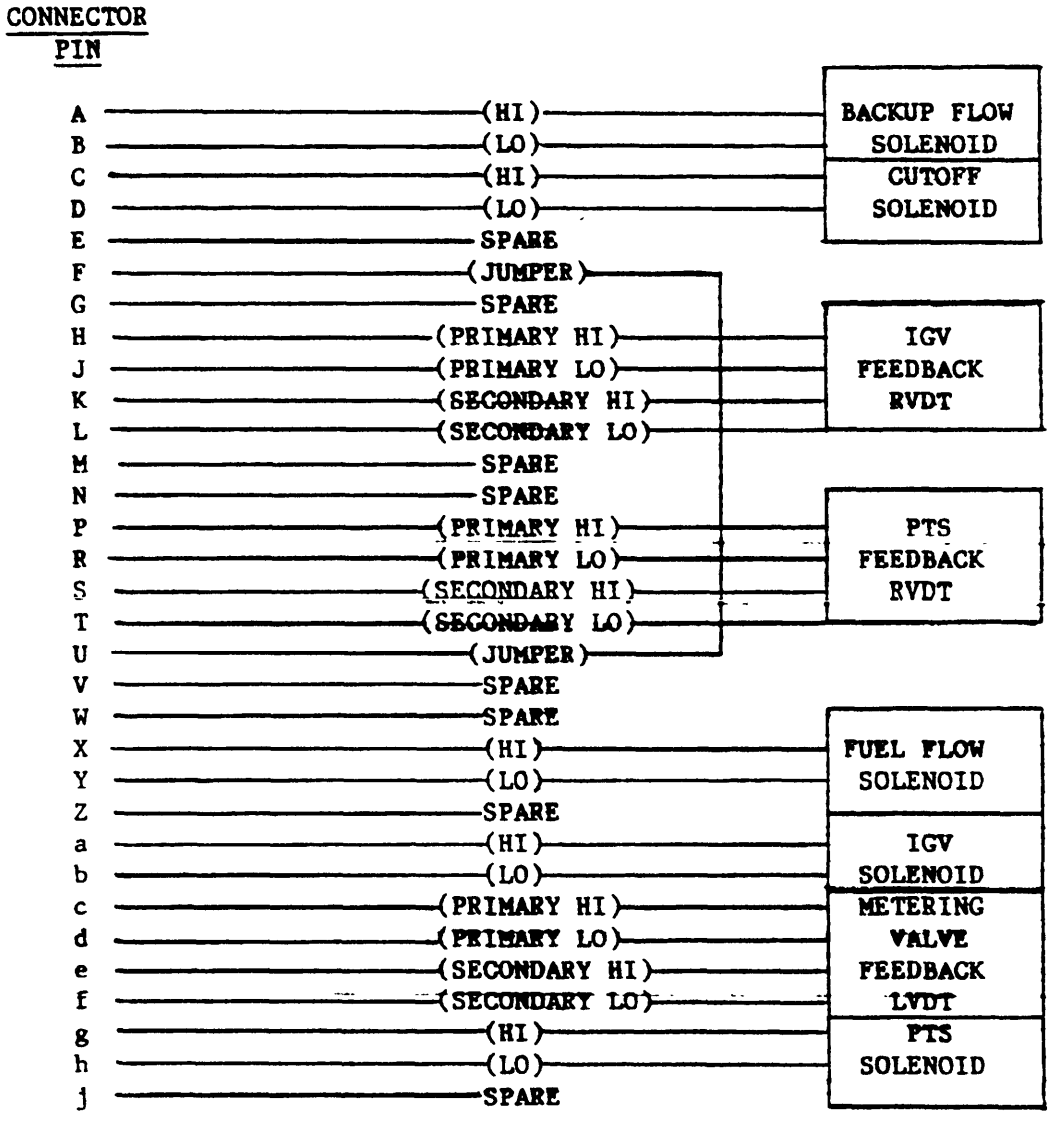


FIGURE 2. Connector pin definition.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.

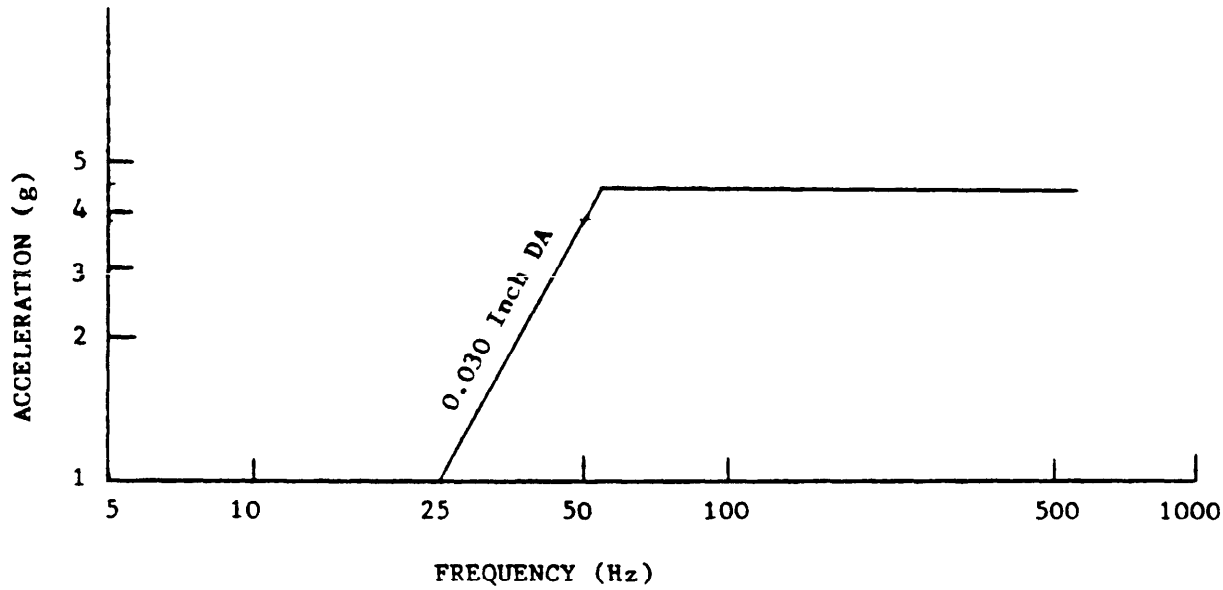


FIGURE 3. Peak vibration envelope.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.

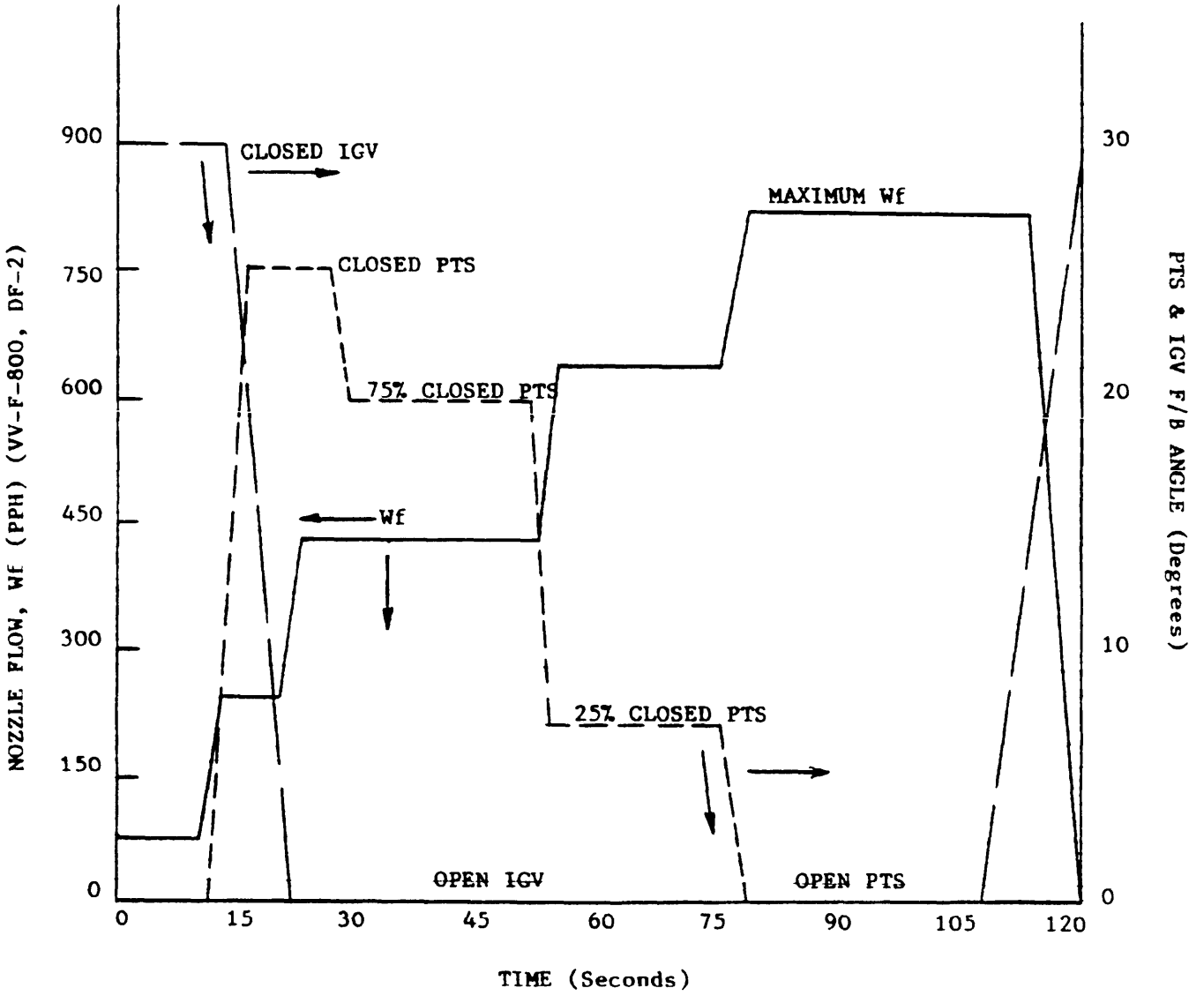


FIGURE 4. Duty cycle.

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:41Z
Check the source to verify that this is the current version before use.

Custodian:
Army - AT

Preparing activity:
Army - AT

(Project 2910-A008)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300
AMSTA–GDS

Commander
US Army Tank–Automotive Command
ATTN: AMSTA–GDS
Warren, MI 48397–5000

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

FSC 2910

DOCUMENT IDENTIFIER AND TITLE

MIL-F-62439(AT); Fuel Control, Main, Electrohydraulic

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:41Z
Check the source to verify that this is the current version before use.